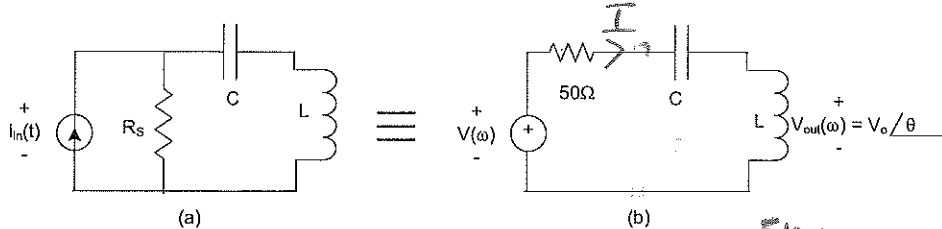


Midterm Exam Solution - Fall 2017

Question 1 [25 Points]:

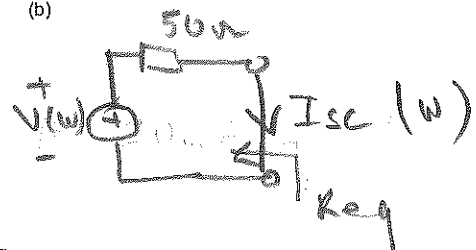
- (a) [4 points] Determine $i_{in}(t)$ and R_S in the circuit given in Figure (a) below, so that it is equivalent to the circuit given in Figure (b), if $V(\omega) = 1 + j0$ V.
- (b) [6 points] If $f = 16$ MHz and $C = 20$ pF, please determine the value of L so that the circuit is at resonance.
- (c) [12 points] Please find V_o and θ if the circuit is resonating at 16 MHz and $C = 20$ pF.
- (d) [3 points] Please write down the corresponding $v_{out}(t)$.



$$V(\omega) = 1 \text{ V}; \quad V(t) = \cos \omega t$$

$$(a) I_{in}(\omega) = I_{sc}(\omega) = \frac{V(\omega)}{50} = \frac{1}{50} = 0.02 \text{ A}$$

$$i_{in}(t) = 20 \text{ mA} \cos \omega t; \quad R_S = R_{eq} = 50 \Omega$$



$$(b) \omega = 2\pi \cdot 16 \times 10^6 \approx 100 \times 10^6 \text{ rad/s}; \quad L = \frac{1}{\omega^2 C} \approx 5 \mu\text{H} \text{ (exact } 4.95 \mu\text{H)}$$

$$(c) I(\omega) = \frac{V(\omega)}{50 \Omega + j\omega L + \frac{1}{j\omega C}}; \quad V_{out}(\omega) = I(\omega) \cdot j\omega L$$

$$\text{at } \omega = 2\pi \cdot 16 \times 10^6 \text{ Hz, } j\omega L + \frac{1}{j\omega C} = 0 \text{ resonance!}$$

$$\therefore I(\omega) = \frac{V(\omega)}{50 \Omega} = \frac{1 \angle 0^\circ}{50 \Omega} = 0.02 \text{ A}$$

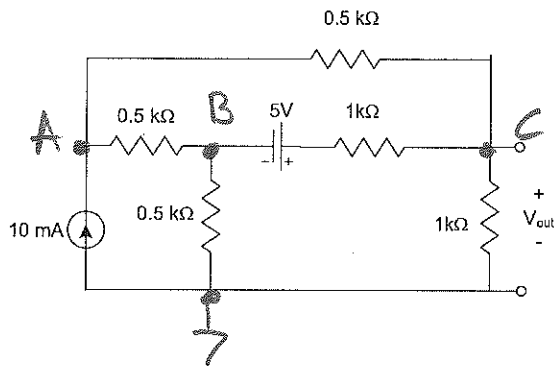
$$V_{out}(\omega) = (j\omega L) I(\omega) \approx j \left[(100 \times 10^6 \text{ rad/s}) (5 \times 10^{-6} \text{ H}) (0.02 \text{ A}) \right] = j 10 \text{ V}$$

$$\therefore V_{out}(\omega) = 10 \angle 90^\circ \text{ V}$$

$$(d) v_{out}(t) = -10 \sin(2\pi \times 16 \times 10^6 t)$$

Question 2 [25 Points]:

Please find V_{out} in the circuit shown below, using Node analysis. Show and explain all your steps.



1. Determine Nodes; 4 Nodes
2. Set Common Node
3. Label Nodes: A, B, C
4. KCL at A, B, C:

$$\text{Node A: } \frac{V_A - V_B}{0.5} + \frac{V_A - V_C}{0.5} - 10 = 0 \Rightarrow 2V_A - V_B - V_C = 5 \text{ V (A)}$$

$$\text{Node B: } \frac{V_B - V_A}{0.5} + \frac{V_B}{0.5} + \frac{V_B + 5 - V_C}{1} = 0 \Rightarrow 2.5V_B - V_A - 0.5V_C = -2.5 \text{ V}$$
$$\text{or } 5V_B - 2V_A - V_C = -5 \text{ V (B)}$$

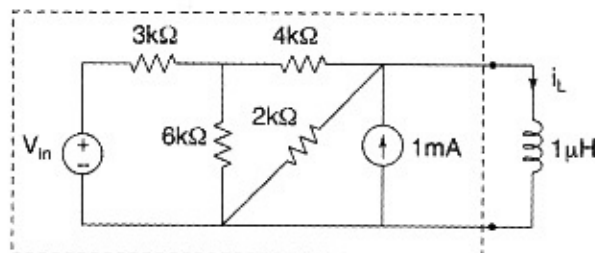
$$\text{Node C: } \frac{V_C - 5 - V_B}{1} + \frac{V_C}{1} + \frac{V_C - V_A}{0.5} = 0 \Rightarrow 2V_C - V_A - 0.5V_B = 2.5 \text{ V}$$
$$4V_C - 2V_A - V_B = 5 \text{ V (C)}$$

Solve A, B, C, simultaneously to find:

$$V_A = 6.25 \text{ V}; V_B = 2.5 \text{ V}; V_C = 5 \text{ V}$$

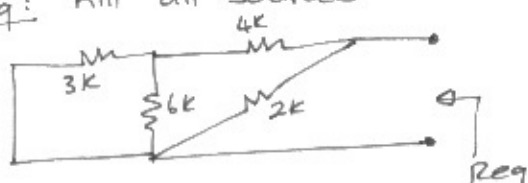
$$\underline{\underline{V_{out} = V_C = 5 \text{ V}}}$$

Question 3 [25 points]:



- (a) [12 points] Find the Norton equivalent of the circuit inside the dashed box, as seen by the inductor. Express the Norton current I_N in terms of V_{in} .
- (b) [5 points] Find the time constant of this first-order RL circuit.
- (c) [8 points] Assume that V_{in} supplies zero voltage before $t=0$, and is switched to $V_{in}=9V$ at $t=0$. Find and plot $i_L(t)$ for all t . **Hint:** Use your result from part (a).

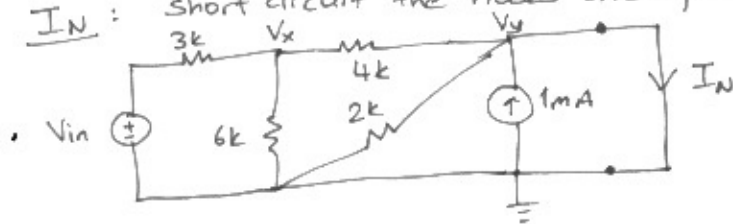
a) Req: Kill all sources



$$R_{eq} = [(3k \parallel 6k) + 4k] \parallel 2k$$

$$= (2k + 4k) \parallel 2k = \boxed{\frac{3}{2} k\Omega}$$

I_N : short circuit the nodes and find the current



$V_y = 0$ due to short circuit path to ground

KCL at V_x :

$$\frac{V_x - V_{in}}{3k} + \frac{V_x}{6k} + \frac{V_x}{4k} = 0$$

(4) (2) (3)

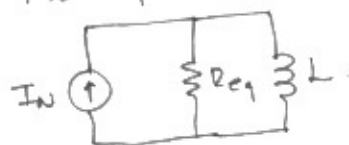
$$9V_x - 4V_{in} = 0 \quad V_x = \frac{4}{9} V_{in}$$

KCL at V_y :

$$\frac{0 - V_x}{4k} + \frac{0 - 0}{2k} - 1mA + I_N = 0$$

$$I_N = 1mA + \frac{V_x}{4k} = \boxed{1mA + \frac{V_{in}}{9k\Omega}}$$

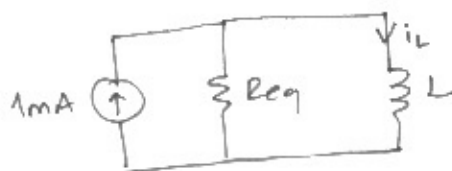
b) From part (a):



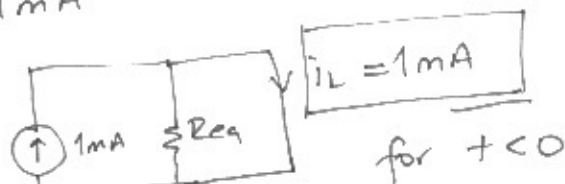
$$\tau = \frac{L}{R_{eq}} = \frac{1 \times 10^{-6}}{\frac{3}{2} \cdot 10^3} = \boxed{\frac{2}{3} ns}$$

c) Before $t=0$, $V_{in}=0$ for a long time. So, inductor must have turned into short circuit.

* During $t < 0$, $I_N = 1mA + \frac{V_{in}}{9k\Omega} = 1mA$



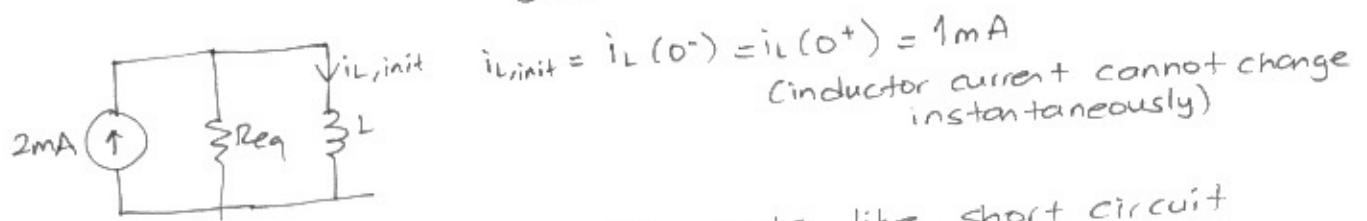
⇒ set inductor to short circuit



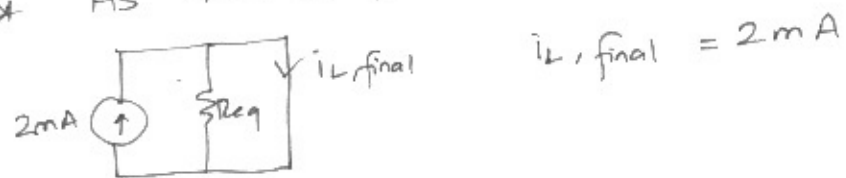
for $t < 0$

* for $t > 0$, $V_{in} = 9V$,

$$I_N = 1mA + \frac{V_{in}}{9k\Omega} = 1mA + 1mA = 2mA$$



* As $t \rightarrow \infty$, inductor acts like short circuit



So, for $t > 0$

$$i_L(t) = i_{L,final} + (i_{L,init} - i_{L,final})e^{-t/\tau}$$

$$= 2mA + (1mA - 2mA)e^{-t/\tau} = 2mA - 1mA \cdot e^{-t/\tau}$$

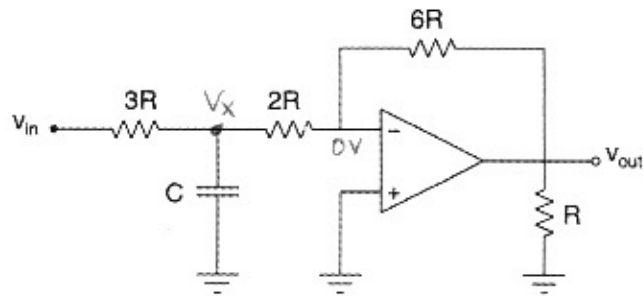
So, overall:

$$i_L(t) = \begin{cases} 1mA, & t < 0 \\ (2 - 1 \cdot e^{-t/\tau})mA, & t \geq 0 \end{cases}$$

where $\tau = \frac{2}{3}ns$



Question 4 [25 Points]:



For the circuit above, $R = 1\text{k}\Omega$, $C = 5\text{ nF}$. Assume v_{out} is not saturated and v_{in} is sinusoidal.

- (a) [10 points] Find the transfer function $H(\omega) = \frac{V_{out}}{V_{in}}$ in terms of ω , R , C .
 (b) [5 points] What kind of filter is this? What is its maximum gain in dB?
 (c) [5 points] What is the 3-dB cut-off frequency in units of Hz?
 (d) [5 points] Find $v_{out}(t)$ if $v_{in}(t) = 5 \cos(\omega_c t + \pi/4)$ Volts, where ω_c is the 3-dB cut-off frequency.

a)

$$V_+ = V_- = 0\text{ V}$$

$$\text{KCL at } V_x: \frac{V_x - V_{in}}{3R} + \frac{V_x - 0}{\frac{1}{j\omega C}} + \frac{V_x - 0}{2R} = 0$$

$$\frac{V_x - V_{in}}{3R} + j\omega C \cdot V_x + \frac{V_x}{2R} = 0$$

(2) (6R) (3)

$$V_x (5 + 6j\omega RC) = 2V_{in}$$

$$\text{KCL at } V_-: \frac{-V_x}{2R} - \frac{V_{out}}{6R} = 0 \Rightarrow V_{out} = -3V_x$$

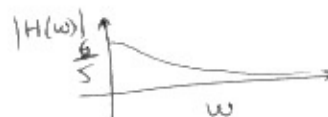
$$V_{out} = -3 \cdot \frac{2V_{in}}{5 + 6j\omega RC} = -\frac{6}{5} \cdot \frac{V_{in}}{1 + \frac{6}{5}j\omega RC}$$

$$H(\omega) = \frac{V_{out}}{V_{in}} = -\frac{6}{5} \cdot \frac{1}{1 + \frac{6}{5}j\omega RC}$$

$$|H(\omega=0)| = 6/5$$

$$|H(\omega \rightarrow \infty)| = 0$$

b) $|H(\omega)| = \frac{6}{5} \cdot \frac{1}{\sqrt{1 + (\frac{6}{5}\omega RC)^2}}$



This is a low-pass filter (LPF).

Maximum $|H(\omega)|$ is at $\omega=0$. $|H(\omega=0)| = \frac{6}{5}$

$$20 \cdot \log_{10} |H(0)| = 20 \cdot \log_{10} \left(\frac{6}{5}\right) = \boxed{1.58 \text{ dB}}$$

$$c) |H(\omega_c)| = \frac{|H(\omega)|_{\max}}{\sqrt{2}} \Rightarrow \frac{6}{5} \cdot \frac{1}{\sqrt{2}} = \frac{6}{5} \cdot \frac{1}{\sqrt{1 + \left(\frac{6}{5}\omega_c RC\right)^2}}$$

$$\Rightarrow \frac{6}{5} \omega_c RC = 1$$

$$\omega_c = \frac{1}{\frac{6}{5} RC}$$

$$f_c = \frac{\omega_c}{2\pi} = \frac{1}{2\pi \cdot \frac{6}{5} \cdot RC} = \frac{1}{2\pi \cdot \frac{6}{5} \cdot 10^3 \cdot 5 \cdot 10^{-9}} = \boxed{26.5 \text{ kHz}}$$

$$d) V_{in} = 5 \cdot e^{j\pi/4}$$

$$H(\omega_c) = -\frac{6}{5} \cdot \frac{1}{1 + \frac{6}{5}j} \cdot \frac{1}{\frac{6}{5} \cdot RC} \cdot RC = -\frac{6}{5} \cdot \frac{1}{1+j}$$

$$V_{out} = V_{in} \cdot H(\omega_c) = -\frac{6}{5} \cdot \frac{5 \cdot e^{j\pi/4}}{1+j} = -\frac{6 \cdot e^{j\pi/4}}{\sqrt{2} \cdot e^{j\pi/4}} = -3\sqrt{2} \\ = 3\sqrt{2} \cdot e^{j\pi}$$

So, $V_{out}(t) = 3\sqrt{2} \cos(\omega_c t + \pi) \text{ V}$
 OR
 $V_{out}(t) = -3\sqrt{2} \cos(\omega_c t) \text{ V}$